

# JKBOSE 2026

## CLASS 11<sup>TH</sup>

Bhaichara  
ON  
TOP

#XUV   
**ORGANIC CHEMISTRY:**  
**SOME BASIC PRINCIPLES**  
**AND TECHNIQUES**

**LECTURE-15**

**CHEMISTRY**

# VISHAL SIR



JKBose Wallah

# ***Topics To Covered***

- 1** **Methods of Purification of Organic Compounds**
- 2** **Qualitative analysis    Part 2**

# REVISION

## METHODS OF PURIFICATION.

- a) SUBLIMATION    b) Crystallisation    c) Distillation
- d) Differential Extraction    e) Chromatography.

# Qualitative And Quantitative Analysis of Organic Compound

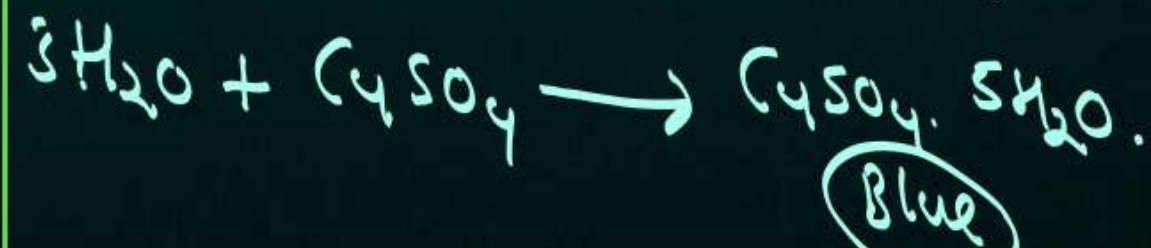
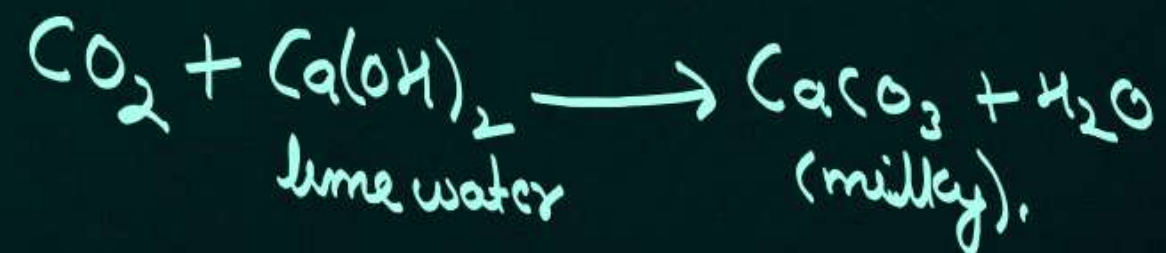
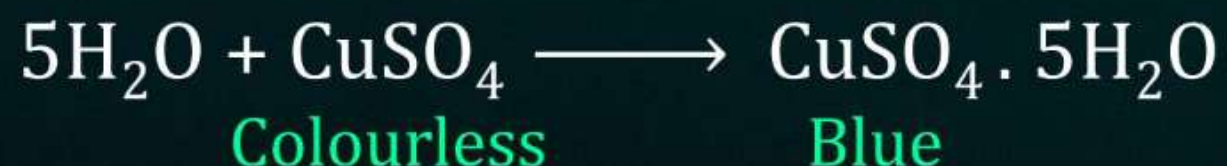


Reactant + Reagent  $\longrightarrow$  Product

**Qualitative analysis:** Detection of element present in compound:

**(i) Detection of C and H:** Compound heated with cupric oxide and produced vapour is passed through lime water or anhydrous  $\text{CuSO}_4$ . If lime water turn milky, it shows presence of  $\text{CO}_2$  or carbon.

If anhydrous  $\text{CuSO}_4$  turn blue than it shows presence of  $\text{H}_2\text{O}$  or hydrogen.,



Unknown  
Chemical  
(organic).



Identify  
Elements

(Qualitative  
Analysis).

a) C & H.

b) ~~X~~, S, P, N.

$X, S, P, N \rightarrow$  Lassaigne's Test.

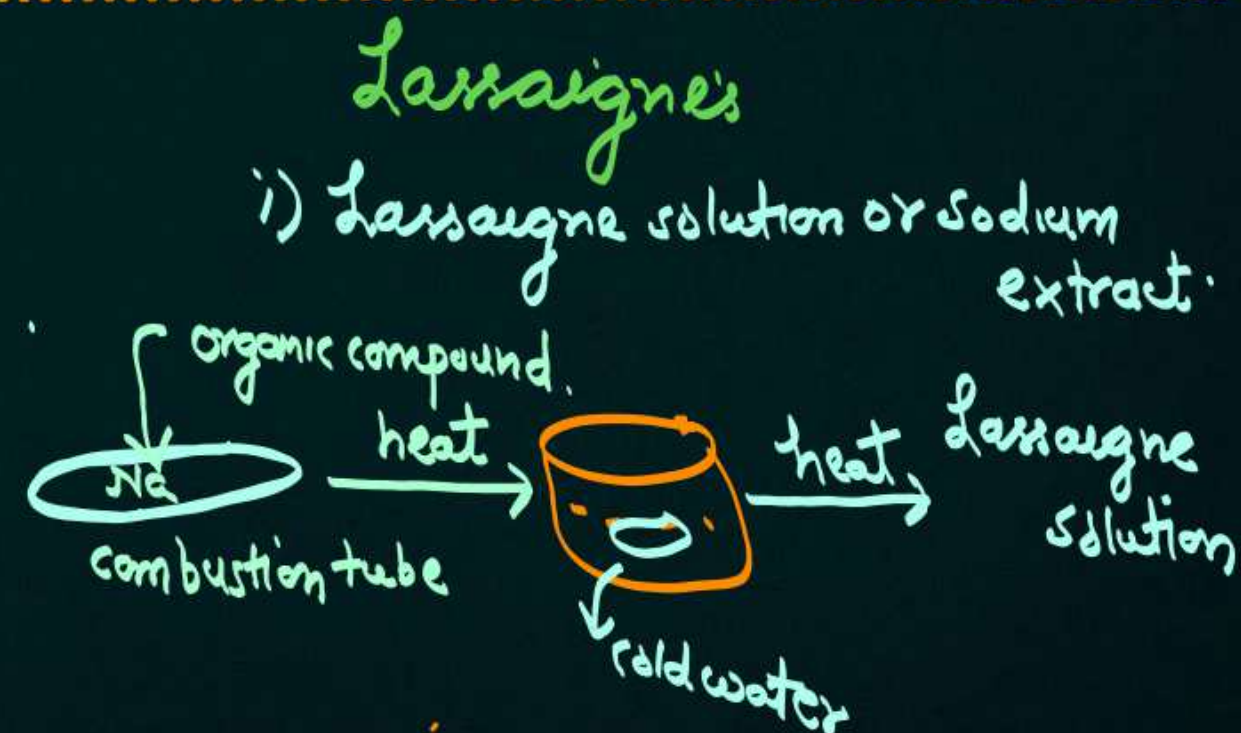


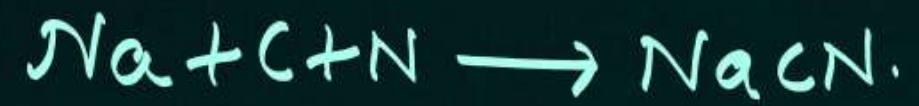
## (ii) Detection of Nitrogen, Sulphur Phosphorous and halogen: [Lassaigne's Test]

### Preparation of lassaigne's solution or sodium extract.

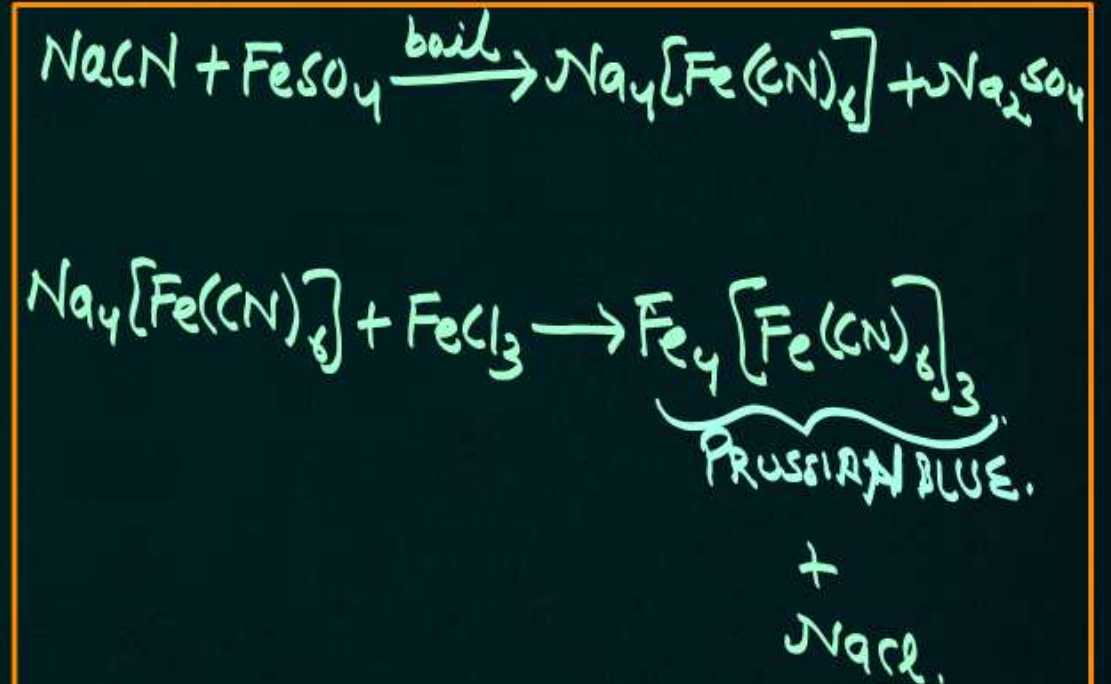
Compound is heated with sodium in combustion tube and poured in cold water and heated for some time it gives lassaigne's solution or sodium extract.

In lassaigne solution elements present in compound get converted from covalent to ionic.





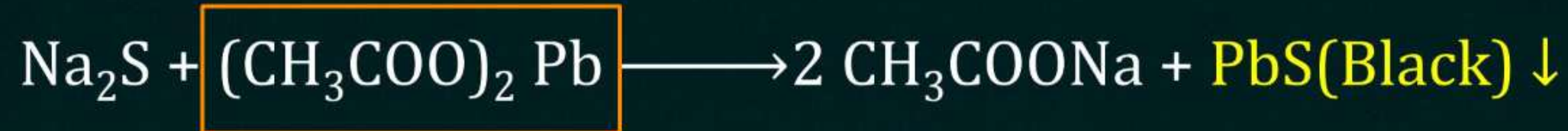
**(a) Test of Nitrogen:** Sodium extract is boiled with  $\text{FeSO}_4$  which gives sodium hexacyano ferrate (II). Now few amount of  $\text{FeCl}_3$  is added which gives prussian blue coloured ferro ferri cyanide that shows presence of Nitrogen.



**(ii) Test of sulphur:** Sulphur is present in form of  $\text{Na}_2\text{S}$

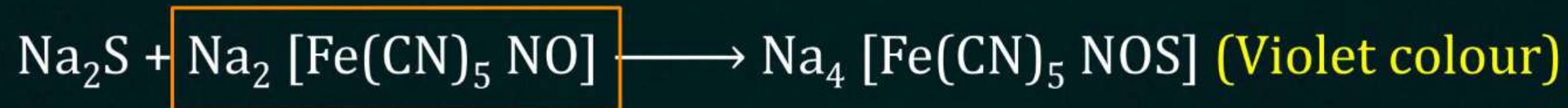
*LEAD ACETATE Test*

**(a) Sodium Extract** is acidified with acetic acid and lead acetate is added to it which gives black ppt of  $\text{PbS}$  and shows the presence of Sulphur

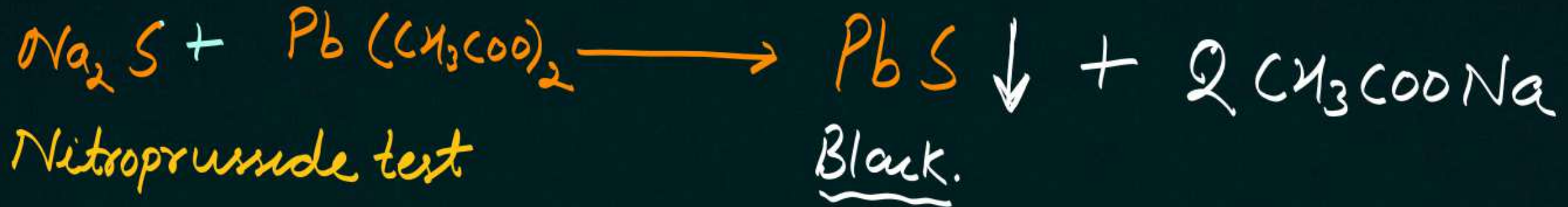


**(b) Nitroprusside test**

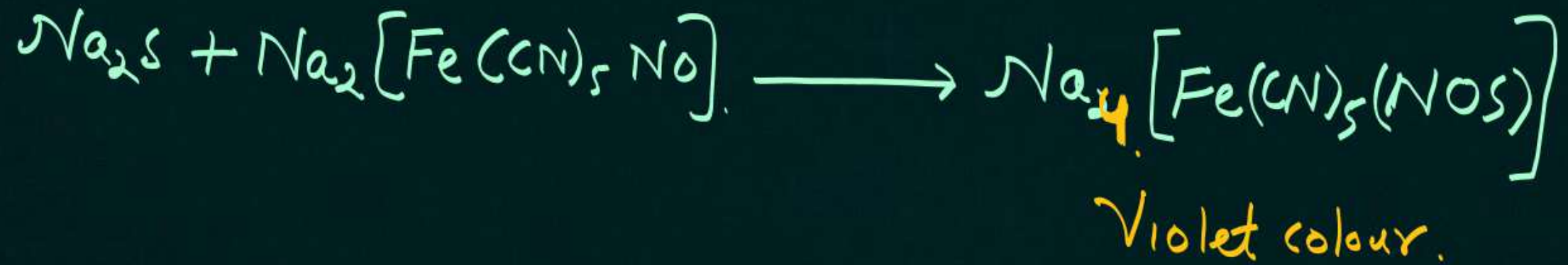
Sodium extract + Sodium Nitroprusside



1) Lead Acetate test



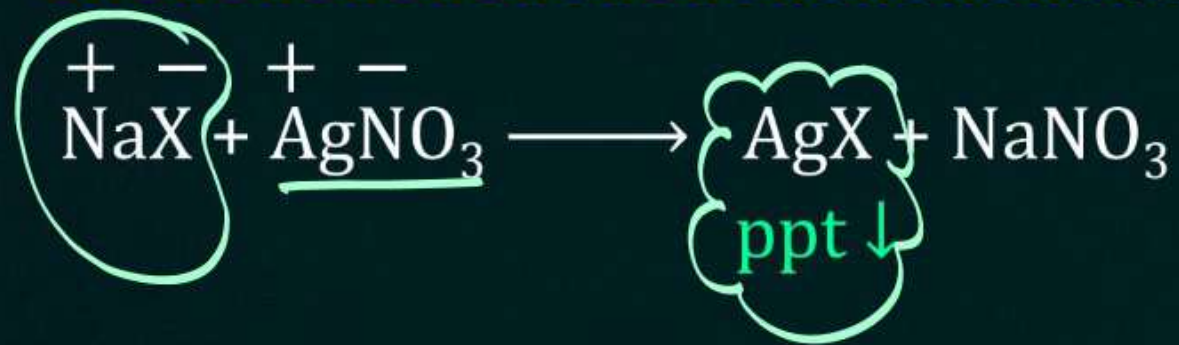
2) Nitroprusside test



**(iii) Test of halogen:** Before the test of halogen sodium extract is boiled with conc.  $\text{HNO}_3$  to decompose the  $\text{Na}_2\text{S}$  and  $\text{NaCN}$  in form of  $\text{H}_2\text{S}$  and  $\text{HCN}$



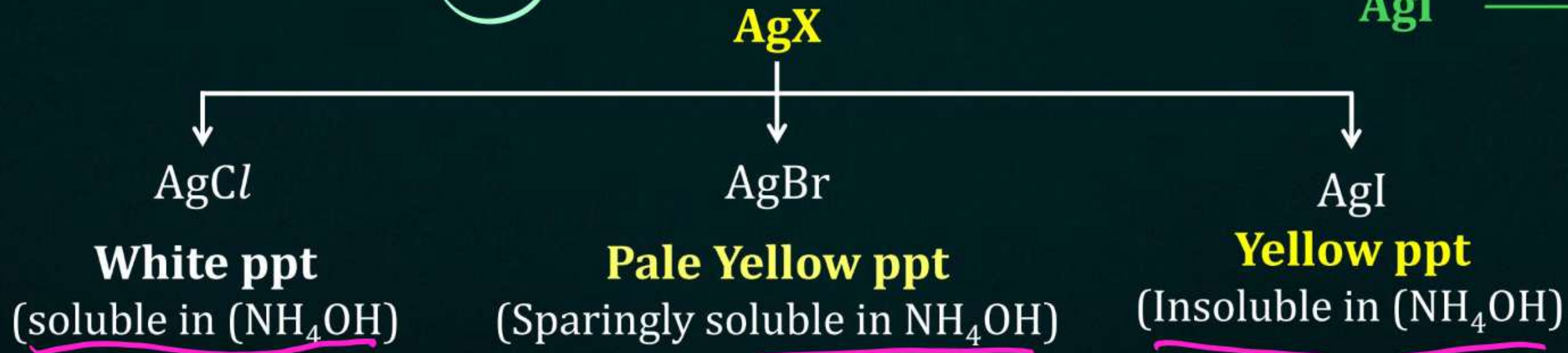
Now sodium extract is treated with silver nitrate which gives precipitate of  $\text{AgX}$ .



$\text{AgCl} \longrightarrow$  white ppt

$\text{AgBr} \longrightarrow$  pale yellow

$\text{AgI} \longrightarrow$  dark yellow

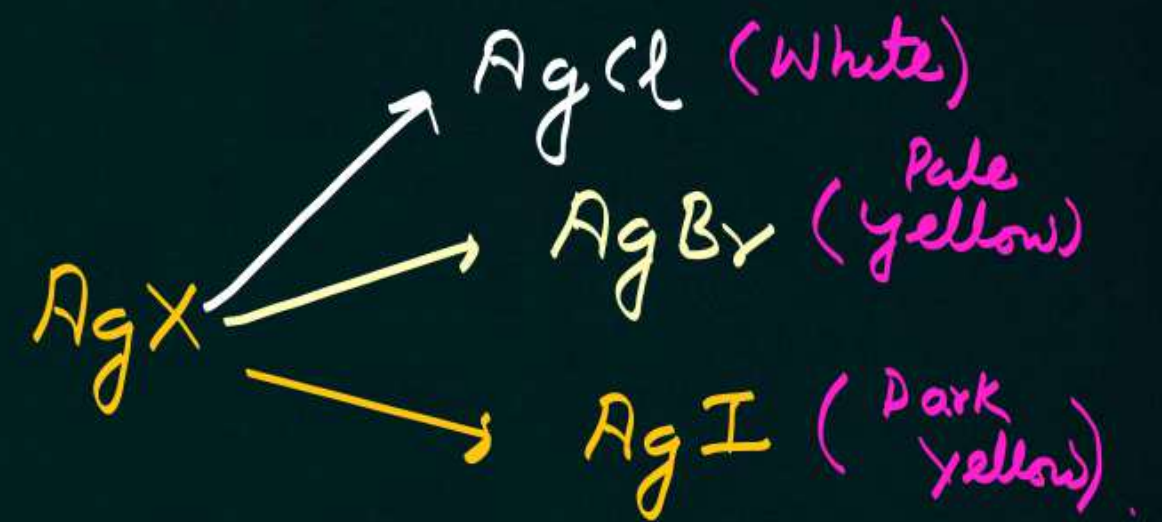


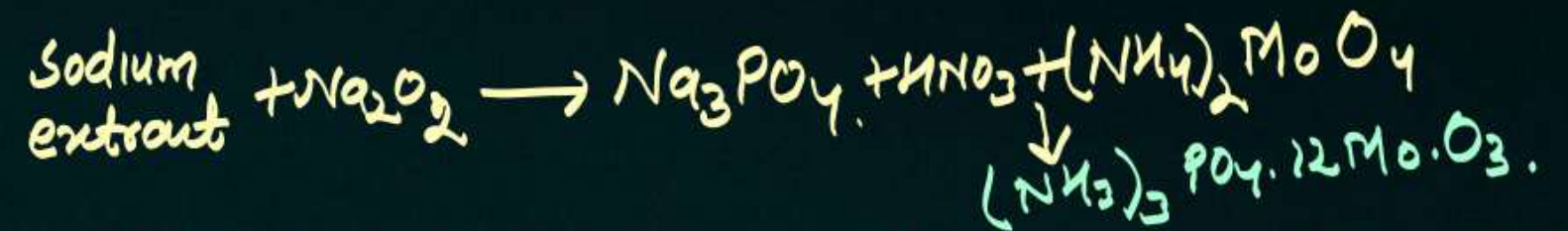
## Test for halogen:→

- Sodium extract +  $\text{HNO}_3$  (boil)  $\begin{cases} \rightarrow \text{H}_2\text{S}^\uparrow \\ \rightarrow \text{HCN}^\uparrow \end{cases}$

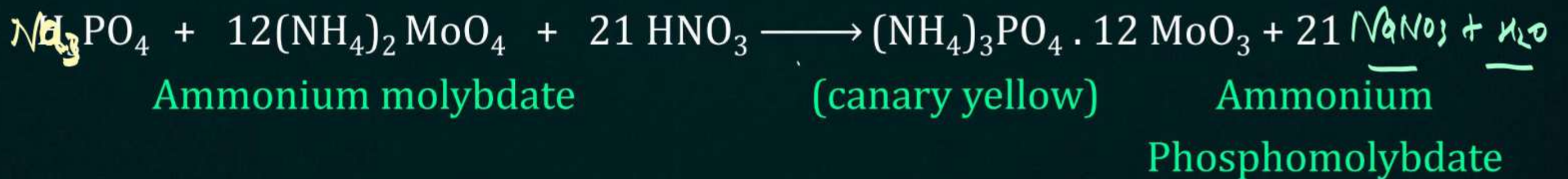


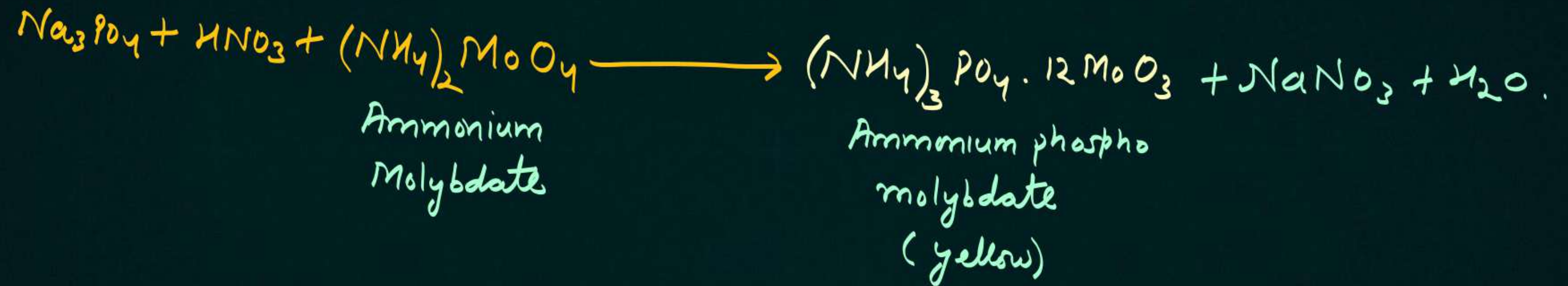
- Sodium extract +  $\text{AgNO}_3 \rightarrow \text{AgX}$  ppt.





**(iv) Test of phosphorous:** Compound is heated with an oxidising agent  $[\text{Na}_2\text{O}_2]$  so the phosphorus in compound is converted in to phosphate which on further reaction with ammonium molybdate in presence of  $\text{HNO}_3$  gives canary yellow ppt of ammonium phosphomolybdate and shows the presence of phosphorus.



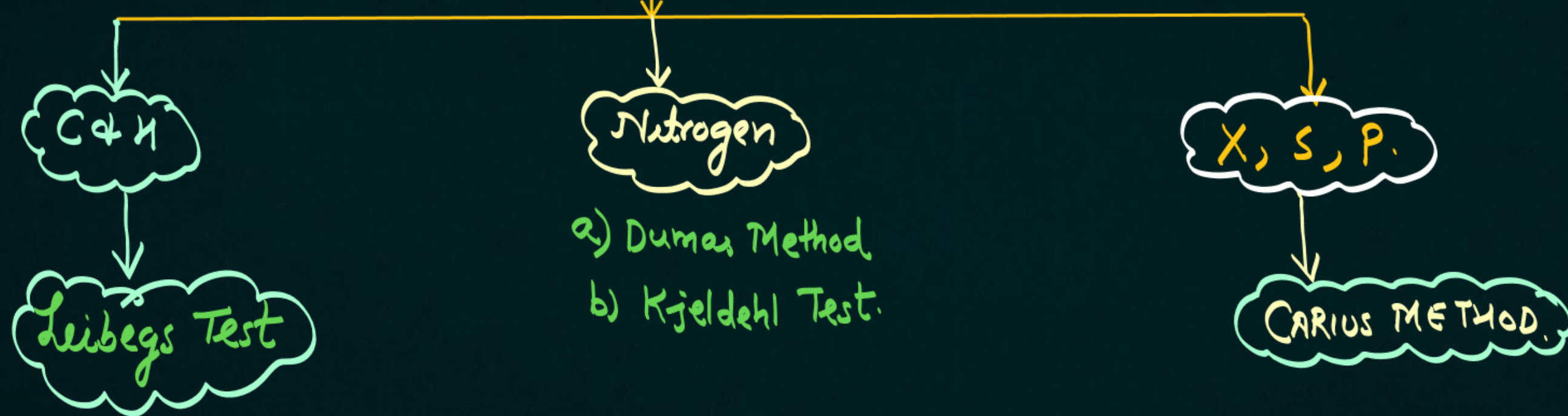


# Qualitative Analysis



# QUANTITATIVE ANALYSIS.

→ Estimation of % composition of any element present in a organic compound.



IUPAC

GOC

METHODS OF PURIFICATION

Qualitative and Quantitative

**QUESTION**

S, X, N, P

Lassaigne's test is used in qualitative analysis to detect

- 1 Nitrogen
- 2 Sulphur
- 3 Chlorine
- 4 All of these

## QUESTION

The sodium extract of an organic compound on acidification with acetic acid and addition of lead acetate solution gives a black precipitate. The organic compound contains.

- 1 Nitrogen
- 2 Halogen
- 3 Sulphur
- 4 Phosphorus

Sodium Extract +  $Pb(CH_3COO)_2 \rightarrow$  Black ppt  
PbS

## QUESTION

The prussian blue colour obtained during the test of nitrogen by Lassaigne's test is due to the formation of:

- 1  $\text{Fe}_4[\text{Fe}(\text{CN})_6]_3$
- 2  $\text{Na}_3 [\text{Fe}(\text{CN})_6]$
- 3  $\text{Fe}(\text{CN})_3$
- 4  $\text{Na}_4(\text{Fe}(\text{CN})_5\text{NOS})$

Thank You JIGAR.

**Thank**  
*You*

I  
#  
♡  
U